



ROOT CAUSE UNLOCKED

Unlock Your *Lyme*

A Complete Functional Medicine Guide to Accurate Testing, Co-Infection Assessment, Drainage-First Recovery, and the Lyme-Thyroid Connection

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

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BEFORE YOU START

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Where the honest boundary sits, before anything else

I want this stated on the way in rather than buried, because it governs every chapter that follows.

Lyme disease is a bacterial infection, and its treatment is antibiotic. That treatment is guided by a joint clinical practice guideline from the Infectious Diseases Society of America, the American Academy of Neurology, and the American College of Rheumatology.¹⁰ Most patients fully recover with recommended therapy.¹¹ Prescription antimicrobials, which class, in what order, and for how long, are decisions for the physician managing that part of your care. They are not something I prescribe, and where this book names them it is so that you know what exists and can raise it with that physician.

No supplement treats the infection. Nothing in this book is offered as an antimicrobial. What the rest of this book is for is different and, I would argue, more useful: understanding why the standard test misses real cases, what a fuller testing picture can and cannot tell you, and the terrain work that determines how well you tolerate treatment and how completely you recover afterward. Sleep, an unrepaired gut, a nervous system held in threat, and an activity pattern that guarantees crashes are the drivers I see left unaddressed for years while people chase the organism. Killing something is not the same as recovering.

Post-treatment Lyme disease syndrome is a genuine condition, and there is no proven treatment for it. Fatigue, musculoskeletal pain, and cognitive complaints persisting six months or longer after adequate antibiotic therapy is a described phenomenon with recognized risk factors, and optimal management awaits a better understanding of the mechanism.¹² A real condition with no proven treatment is exactly the space where an industry grows, and the burden of proof sits on anyone selling into it, including me. Persistent symptoms are not a reason for indefinite antibiotics, which is an infectious disease physician's decision, and not a reason to buy an eradication protocol either.

One instruction here is not optional: tell your physician everything you are taking. People frequently run herbal protocols alongside prescription antibiotics. Herbs are pharmacologically active, some affect the same liver enzymes that clear medications, and some affect bleeding. Bring the actual labels, not the product names.

The Test Said Negative. The Body Tells a Different Story.

Lyme disease is one of the most contested diagnoses in modern medicine, and the controversy is more institutional than scientific. The science on *Borrelia burgdorferi*, on tick-borne co-infections, and on the limitations of the standard two-tier Lyme test is substantially more settled than the public clinical conversation suggests. The patients caught in the middle of this controversy are those who present with a constellation of symptoms that fits no other diagnosis, who have been exposed to tick-endemic environments, whose standard tests are negative, and who are told by successive providers that their symptoms are unexplained, stress-related, or psychiatric.

The standard Western Blot test was designed with epidemiological specificity in mind: to minimize false positives in the general population during acute illness. For this purpose, it performs adequately. For chronic or late-disseminated Lyme disease, it fails a clinically significant percentage of real cases. The test analyzes a limited panel of *Borrelia* antigens. It tests nothing for the co-infections that are transmitted by the same ticks and that are present in a substantial proportion of chronic Lyme patients. And it was standardized using serum from patients in the acute stage of illness, a stage at which antibody profiles differ substantially from those of patients with late or persistent infection.

What makes chronic tick-borne illness clinically complex is the co-infection picture, more than the *Borrelia* organism itself. *Babesia*, *Bartonella*, *Ehrlichia*, and *Anaplasma* are transmitted by the same *Ixodes* ticks. Each produces a distinct clinical syndrome. Each requires different treatment approaches. A patient infected with both *Borrelia* and *Babesia* who is treated only for *Borrelia* may see limited improvement, because the organism driving their most disabling symptoms, the profound fatigue, the air hunger, the cyclical flares, is a different organism requiring a different drug class from their physician. In my clinical experience, incomplete pathogen identification is one of the more consistent explanations when Lyme treatment does not deliver, and I want to be careful with that claim rather than sweeping: it is a pattern I see, not a proven account of why every treatment fails, and it is not the only explanation on the table.

This guide walks through eight keys to understanding and addressing tick-borne illness from the ground up. It covers why the standard test fails, what comprehensive testing actually reveals, how co-infections change the clinical picture, the role of the gut in both Lyme pathophysiology and Lyme recovery, the drainage-first framework that prevents treatment-induced crises, and the often-missed connection between Lyme disease and Hashimoto's thyroiditis that affects a meaningful proportion of chronic Lyme patients. Read it as a framework for understanding your own clinical picture.

Why the Standard Test Fails

The two-tier testing protocol recommended by the Centers for Disease Control for Lyme disease consists of an ELISA screening test followed by a Western Blot confirmation test in patients who screen positive. This protocol was designed in the 1990s for a specific purpose: surveillance and outbreak detection in populations with high background exposure. Its application as a clinical diagnostic tool for individual patients, particularly those with chronic symptoms and possible late-disseminated or persistent infection, has been the source of ongoing clinical controversy precisely because the test's design parameters do not match that clinical use case.

The Sensitivity Problem

The ELISA screening test that begins the two-tier protocol has documented sensitivity of approximately 30 to 40 percent for early Lyme disease.¹ For late-disseminated Lyme, sensitivity estimates in independent research vary considerably depending on the study population and the elapsed time since infection, and a meaningful share of patients with confirmed late Lyme disease will have a negative ELISA and will never proceed to Western Blot confirmation under the standard two-tier protocol. They receive a negative test result and are sent home.

The Western Blot confirmation test compounds this problem. The CDC surveillance criteria for a positive Western Blot require the presence of 5 of 10 specific immunoglobulin G bands for IgG, or 2 of 3 specific bands for IgM. These criteria were designed to maximize specificity for surveillance purposes. In individual clinical diagnosis, they exclude patients with partial antibody responses, patients whose immune profiles have shifted over time, and patients infected with *Borrelia* strains not fully represented in the original band selection criteria. The standard Western Blot tests 10 *Borrelia* antigens. The Vibrant Tickborne 2.0 PCR panel tests 41.

The Co-Infection Blind Spot

The most significant limitation of the standard Lyme test lies in what it leaves out entirely, more than in its sensitivity for *Borrelia*. The standard two-tier protocol tests for *Borrelia burgdorferi* and nothing else. It does not test for *Babesia*, *Bartonella*, *Ehrlichia*, or *Anaplasma*. These organisms are transmitted by the same Ixodes ticks that transmit *Borrelia*, and they are present in a clinically significant proportion of tick-borne illness patients.

Babesia microti and *Babesia duncani* are intraerythrocytic parasites, organisms that infect red blood cells in a manner analogous to malaria. They produce a clinical syndrome characterized by profound cyclical fatigue, night sweats, air hunger, headaches, and fevers that does not respond to standard Lyme antibiotics because antiprotozoal agents, not antibiotics, are required to treat *Babesia*. A patient with both *Borrelia* and *Babesia*

who is treated only with antibiotics targeted at *Borrelia* will experience partial improvement at best, because the organism most responsible for their fatigue and air hunger is not responsive to the treatment protocol.

Bartonella henselae and *Bartonella quintana* produce a neurological and psychiatric symptom profile that is among the most commonly misattributed in all of medicine. Tingling and burning sensations, anxiety, insomnia, mood instability, brain fog, cognitive dysfunction, and OCD-like intrusive thoughts are classic *Bartonella* presentations that are routinely attributed to anxiety disorders or psychiatric conditions. Patients with active *Bartonella* frequently spend years in mental health treatment before the tick-borne connection is identified. Standard Lyme panels miss *Bartonella* entirely.

Ehrlichia and *Anaplasma* are obligate intracellular bacteria that infect white blood cells. They produce an acute febrile illness that can be severe but is generally responsive to doxycycline.² In chronic presentations, they produce a pattern of persistent immune dysfunction and inflammatory burden that is clinically important to identify because it affects treatment sequencing and duration.

The Co-Infection Picture

Understanding the specific clinical presentations of each major tick-borne co-infection is essential for interpreting symptoms accurately and for understanding why a Lyme treatment protocol that does not address co-infections produces limited results. The following descriptions are selective rather than exhaustive, capturing the clinical features most relevant to patients with chronic tick-borne illness.

Borrelia Burgdorferi: The Primary Lyme Organism

Borrelia burgdorferi is a spirochete, a corkscrew-shaped bacterium with a remarkable capacity for immune evasion. It can alter its surface proteins to evade antibody recognition, invade intracellular environments where antibiotics penetrate poorly, and form biofilm-protected aggregates that resist standard antibiotic concentrations. These adaptations explain why short-course antibiotic treatment produces lasting resolution in early Lyme but frequently fails to eliminate the organism in late-disseminated infection.

The classic bull's-eye rash, erythema migrans, appears in most early Lyme infections but is frequently not observed because it appears at the site of the tick bite, which may be in an area the patient cannot easily examine. The absence of a rash does not rule out *Borrelia* infection. The rash, when present, may also appear as a non-classic red oval or expanding redness without the central clearing that produces the classic target appearance.

Chronic *Borrelia* infection produces the constellation most associated with Lyme disease in the public understanding: migratory joint pain, neurological symptoms including brain fog and cognitive impairment, fatigue that is disproportionate to activity level, and sleep disturbances. The migratory quality of the joint pain, moving from location to location over days to weeks, is particularly characteristic and helps distinguish it from inflammatory arthritides that tend to affect consistent locations.

Babesia: The Overlooked Parasite

Babesia is arguably the most clinically disabling of the tick-borne co-infections in terms of its effect on daily function. The organism infects red blood cells and reproduces within them, periodically releasing new organisms in cycles that produce the characteristic symptom pattern of cyclical episodes of severe fatigue, drenching night sweats, air hunger, low-grade fever, headaches, and emotional lability. Between cycles, patients may feel relatively functional. During cycles, they may be profoundly debilitated.

Air hunger, the sensation of being unable to take a satisfying breath despite normal oxygen saturation, is highly characteristic of *Babesia* and helps distinguish it from *Borrelia*-dominant presentations. Patients often describe feeling as though they are breathing through a wet blanket or cannot fully fill their lungs. This symptom is often attributed to anxiety. When it co-occurs with cyclical fatigue and night sweats, *Babesia* should be in the differential regardless of standard Lyme test results.

Babesia is treated with antiprotozoal combinations, typically atovaquone and azithromycin, rather than the doxycycline and amoxicillin regimens used for Borrelia.³ Those are prescription antimicrobials that physicians use for babesiosis. They are not something I prescribe, and they are named here so you know the distinction exists and can raise it with the physician managing that part of your care. The practical point for you is simply that treating one organism does not treat the other, so a patient carrying active Babesia while being treated for Borrelia may improve in the symptoms Borrelia drives and stay disabled in the ones Babesia drives.

Bartonella: The Neuropsychiatric Infection

Bartonella is transmitted both by ticks and by cat scratch (from fleas carrying the organism). In the context of tick-borne illness, Bartonella henselae and Bartonella quintana are the primary species. Bartonella infects endothelial cells lining blood vessels and produces a neurological and psychiatric symptom profile that is unique among the tick-borne co-infections.

The neurological signature of Bartonella includes tingling, burning, or electric sensations in the extremities and across the skin surface that patients often describe as “fizzing” or feeling like they are being mildly electrocuted. Profound anxiety, sometimes approaching panic disorder severity, is extremely common. Intrusive, repetitive, or OCD-like thoughts that emerged or worsened with illness onset are a Bartonella signature that most psychiatrists have not been trained to recognize as infectious in origin. Insomnia, particularly difficulty maintaining sleep, is near-universal in active Bartonella infection.

The clinical significance of recognizing Bartonella is substantial, because a patient with active Bartonella who is in psychiatric treatment for anxiety or OCD may be receiving symptomatic management while an infectious contributor goes unexamined. The proposed mechanism is that these are neurological consequences of an organism with tropism for central nervous system vasculature rather than a primary psychiatric disorder, and that model is a reason to look, not an established account of every case. Two things follow, and they matter more than the mechanism. If Bartonella is present and treatable, treating it is a physician's decision and worth pursuing on its own terms. And nothing about that is a reason to stop psychiatric treatment or a prescribed psychiatric medication, which is a conversation with the prescriber and never a decision to make from a book.

Ehrlichia and Anaplasma

Ehrlichia and Anaplasma are obligate intracellular bacteria transmitted by ticks that infect white blood cells. They produce an acute febrile syndrome with headache, myalgia, leukopenia, thrombocytopenia, and elevated liver enzymes. In acute presentation, they are generally recognized and treated with doxycycline. In chronic presentations, they contribute to a persistent state of immune dysfunction characterized by impaired neutrophil function, which reduces the immune system's capacity to control Borrelia and other co-infections. Identifying and treating Ehrlichia and Anaplasma is important as part of the complete co-infection picture because their effect on immune function affects the clinical response to Borrelia treatment.

The Right Testing Framework

Accurate pathogen identification is what lets the rest of a plan be built on data rather than on assumption. The organisms present, their relative burden levels, and the co-infection combination are inputs to a prescribing decision that belongs to your physician: which antimicrobial agents are appropriate, in what order, and for how long is that physician's call, not mine and not a book's. What I do with the same information is different and complementary. It tells me which terrain work matters most for you, whether mold or gut or thyroid needs to run alongside, and how much preparation your drainage capacity is going to need before treatment starts. Testing that provides an incomplete picture leaves both of those decisions guessing. The comprehensive testing framework I use at Root Health is designed to characterize the full tick-borne burden early, so that whoever is prescribing has the fullest picture available to them.

The Vibrant Tickborne 2.0 PCR Panel

The Vibrant Tickborne 2.0 PCR panel represents the most comprehensive commercially available tick-borne testing. It uses polymerase chain reaction technology to detect the DNA of tick-borne organisms directly, rather than relying solely on the antibody response, which can be diminished in chronic infection or in patients with immune dysregulation. The PCR approach detects active infection even when the antibody response is insufficient to produce a positive result on standard serology-based tests.

The catch is that PCR detects DNA in circulating blood. Spirochetes and other tick-borne organisms that have migrated into joint fluid, connective tissue, or central nervous system compartments may not be detectable in a blood draw at any given moment. A negative PCR result does not rule out infection when symptom burden is clinically consistent with tick-borne illness. In those cases, the full clinical picture, including symptom pattern, exposure history, and response to empirical support, informs the decision-making alongside the laboratory data.

The panel tests for *Borrelia burgdorferi* across 41 antigens, compared to the 10 on a standard Western Blot. The significantly expanded antigen coverage means that *Borrelia* strains and antibody patterns that would produce a negative or indeterminate result on standard testing may produce a positive result on the Tickborne 2.0 panel. It tests for *Babesia microti* and *Babesia duncani* separately, recognizing that the two species have different geographic distributions and potentially different clinical profiles. It tests for *Bartonella henselae* and *Bartonella quintana*. It covers *Ehrlichia* and *Anaplasma* species, tick-borne encephalitis, and relapsing fever organisms. It provides the breadth of coverage necessary to build a fuller clinical picture rather than a partial one. It is available through rootcauseunlocked.com/order-labs.

And the honest caveat on the panel as a whole. This is a specialty panel, not a validated replacement for guideline serology, and I am not going to sell it as one. A result on it is read as consistent with a clinical picture, alongside your symptom pattern, your exposure history, and your standard testing. It does not confirm an infection by itself and it does not overrule a physician's diagnosis. There is a documented market

of unorthodox testing and treatment aimed at people in exactly your position,¹³ so apply the same filters to this panel that you would apply to anything else offered to you. Ask whether the assay is validated for this use, who profits from the result, and, if it comes back positive, what specifically changes about your care.

Complementary Panels

Beyond the Tickborne 2.0 panel, several complementary assessments provide context that affects treatment decisions. The Total Tox Burden panel from Vibrant Wellness characterizes the mycotoxin, heavy metal, and environmental chemical burden that coexists with Lyme disease in a high proportion of cases. Read its urinary mycotoxin component as documenting recent exposure rather than as measuring stored body burden, because that is what a urinary panel is able to say. Mold toxin exposure and Lyme infection are not coincidentally correlated. Mycotoxins suppress immune function, impairing the immune system's capacity to control *Borrelia* and co-infections. The inflammatory burden of active Lyme also impairs the liver's capacity to process mycotoxins. The two conditions maintain and amplify each other. Identifying the mold burden determines whether antimold and drainage interventions need to run alongside the antimicrobial protocol.

The Organic Acids Test provides the neurochemical picture, covering neurotransmitter metabolite status, mitochondrial function markers, and fungal overgrowth indicators that collectively characterize how the tick-borne burden has affected the brain's biochemical environment. For patients with significant neurological or psychiatric symptoms, the OAT provides the substrate data that guides neurological support within the broader protocol.

For patients with thyroid symptoms alongside tick-borne illness, the Vibrant Hormone Zoomer establishes the full thyroid picture including TPO and thyroglobulin antibodies. The Lyme-Hashimoto's connection is covered in detail in Key 7. The clinical decision to add thyroid testing should be made for any Lyme patient with fatigue, weight changes, brain fog, cold intolerance, hair thinning, or mood instability that has persisted despite Lyme-focused treatment.

Panel	What It Reveals
Vibrant Tickborne 2.0 PCR	Borrelia (41 antigens), Babesia microti and duncani, Bartonella, Ehrlichia, Anaplasma, tick-borne encephalitis
Total Tox Burden	Mycotoxins (29), heavy metals, environmental chemicals. Co-burden that impairs immune function and recovery.
Organic Acids Test (OAT)	Neurotransmitter metabolites, mitochondrial markers, fungal overgrowth. Neurological biochemistry.
Vibrant Hormone Zoomer	Full thyroid panel, TPO/TgAb antibodies, adrenal cortisol, sex hormones. For Lyme-thyroid assessment.
Vibrant Gut Zoomer Complete	Gut ecosystem, permeability markers, secretory IgA, pathogens. Establishes gut picture before antimicrobials.

The Gut-Lyme Connection

The gut is central to Lyme disease recovery, not incidental to it. The relationship between gut health and Lyme disease operates through several specific mechanisms, and each one affects both the course of the infection and the effectiveness of the treatment protocol. Patients who address Lyme without addressing the gut consistently produce worse outcomes than those whose protocols treat both simultaneously.

How Lyme Compromises the Gut

Borrelia burgdorferi and its co-infections produce systemic immune activation that has specific consequences for gut function. Chronic immune activation driven by tick-borne organisms suppresses secretory IgA production, the gut's primary mucosal immune defense. Secretory IgA normally monitors the gut's microbial inhabitants and prevents pathogenic organisms from establishing colonization. When secretory IgA is depleted, the gut becomes susceptible to pathogenic overgrowth and dysbiosis.

The immune dysregulation of Lyme disease also affects the composition of the gut microbiome directly. Th17-dominant immune polarization, which characterizes both active Lyme and Hashimoto's, promotes the growth of specific bacterial populations that generate more inflammatory LPS burden and further suppresses the commensal organisms that maintain gut barrier integrity. The result is a gut microbiome that, in the presence of active Lyme infection, is progressively less capable of maintaining the gut barrier that protects against systemic antigen exposure.

Antibiotic treatment for Lyme disease compounds this problem. Extended antibiotic courses, which are frequently required for late-disseminated Lyme, produce significant microbiome disruption. The diversity depletion from antibiotics can take twelve to twenty-four months to recover without active microbiome restoration. Patients who complete antibiotic courses without microbiome restoration support emerge with a gut that is both depleted of beneficial organisms and vulnerable to recolonization by pathogenic ones.

How Gut Dysfunction Impairs Lyme Recovery

The bidirectionality of this relationship is clinically critical. A compromised gut barrier generates LPS endotoxemia that sustains the systemic inflammatory burden independently of the Lyme infection itself. Even if antimicrobial treatment effectively reduces the *Borrelia* and co-infection burden, the inflammatory signaling from a leaky gut maintains the same immune activation state that impairs neurological function, drives fatigue, and sustains the Th17 polarization that promotes autoimmune consequences. Patients who have completed Lyme treatment and still feel unwell⁴ frequently have persistent gut dysfunction as a primary driver of their ongoing symptoms.

The liver's capacity to process the microbial debris generated during antimicrobial treatment depends on an adequate supply of glutathione, methylation capacity, and bile acid production. All three of these are impaired

by gut dysfunction. A compromised gut produces less bile acid recycling, depletes glutathione through ongoing oxidative stress, and generates the methyl-depleting inflammatory burden that reduces Phase 2 liver detoxification capacity. The patient undergoing Lyme treatment with a compromised gut is attempting to process a high microbial debris load through a detoxification system that is already overwhelmed.

This is the mechanism usually offered to explain the severity of Herxheimer reactions in Lyme treatment: the drainage and detoxification systems cannot handle the load the antimicrobial is generating. It is a plausible model and it is the one my sequencing assumes, but it remains a model rather than a proven account of what is happening in any individual patient. What I will claim is narrower and more useful. Preparing the gut, restoring gut barrier function, and establishing drainage capacity before antimicrobials are introduced is what I do, because in my clinical experience patients tolerate treatment better on that footing. The full caution about Herxheimer framing is in Key 5, and it is worth reading before you interpret anything you feel in the first weeks of treatment.

The Gut Workup for Lyme Patients

For Lyme patients with significant gut symptoms, the Gut Zoomer Complete provides the comprehensive microbial picture worth having before antimicrobial treatment begins. The secretory IgA result is particularly informative: severely depleted secretory IgA in the context of Lyme disease is consistent with the expected gut immune suppression and is a signal that microbiome restoration support should accompany any antimicrobial protocol. For patients without prominent gut symptoms, the Wheat Zoomer is a minimum assessment of whether gut permeability is contributing to the ongoing inflammatory picture. Both are read as suggestive rather than diagnostic, and the companion Gut kit covers the validation limits of each marker in detail, including why no single number on either panel establishes a leaky gut on its own.

Drainage First

The drainage-first principle is not a theoretical preliminary. In Lyme disease specifically, it is where I put the preparation time, because in my clinical experience it makes the largest difference to whether treatment is tolerated well enough to finish. Patients who start intensive antimicrobial treatment on a depleted, congested baseline are the ones I see stop their protocols, and providers watching that happen sometimes conclude that further intervention is futile when what actually failed was the preparation.

The proposed biological mechanism is straightforward. When *Borrelia* and co-infections are targeted by antimicrobial agents, the organisms die and release their contents into systemic circulation. Spirochete cell wall components, bacterial DNA, endotoxins, and the inflammatory cytokines generated by the dying organisms create an acute burden on the lymphatic system, the liver, and the kidneys. If these systems are not prepared to process that load, the model holds that it accumulates and circulates, producing the Herxheimer reaction: a worsening of symptoms in the days following treatment.⁵

And here is the caution that has to sit alongside that, because it matters more than the mechanism. If feeling better proves the treatment is working and feeling worse also proves it is working, the claim cannot be tested against reality, and an untestable claim is a poor reason to keep going or to keep paying. So I treat worsening as information rather than as a milestone. Sometimes it is what the paragraph above describes and it passes. Sometimes it means the dose is too high, the groundwork was skipped, or this is the wrong plan for you. And sometimes, in a bacterial infection that can affect the nerves, heart, and joints, worsening is the illness itself progressing and needs a physician the same day rather than reassurance from a protocol. Significant worsening means reassess with me, and urgent symptoms mean urgent care. It is never a badge.

I am aware this sits differently from how die-off is usually sold in this space, and that the companion supplement guide in this kit declines to recommend protocols built on Herxheimer framing. That is the same position, stated twice. I keep the drainage sequencing because I think the preparation is worth the time. I do not ask you to read your symptoms as proof of anything.

In patients with chronic Lyme disease, all three drainage systems are typically compromised before treatment even begins. The liver has been processing the inflammatory burden of active infection for months or years. Glutathione is depleted. Methylation capacity is reduced by the ongoing inflammatory burden. The lymphatic system is congested from years of chronic inflammation. The kidneys are managing an elevated toxic metabolite load. Introducing intensive antimicrobial treatment into this environment without first restoring drainage capacity is a predictable setup for treatment failure.

The Drainage Protocol for Lyme

What you are buying here, stated plainly. The drainage sequence below has mechanistic reasoning and clinical use behind it, and no human outcome trials I could find in Lyme disease. I searched for them. That is

what happens to compounds nobody can patent: the trial that would settle the question never gets funded, so absence of evidence here is usually absence of a sponsor rather than absence of an effect. It is also not a licence to assume it works. So this is reasoned clinical practice rather than trial-proven certainty, and I would rather you know which one you are following. None of it is an antimicrobial, none of it treats the infection, and none of it substitutes for the antibiotic treatment your physician prescribes. Three consequences worth acting on. Put your money and your attention into the prescribed treatment, the exposure and mold questions, sleep, and pacing first, because that is where the leverage lies. Hold this protocol to a defined endpoint rather than running it indefinitely. And tell your prescriber every item in it, by label rather than by product name.

Product	Clinical Role	Dosing
Lymph-Tone I (Energetix)	The acute-presentation choice, for symptoms present less than 12 weeks.	Per label direction
Lymph-Tone II (Energetix)	Primary lymphatic and ECM drainage for the majority of Lyme cases. ECM congestion from years of chronic inflammatory burden is universal in late-disseminated Lyme. Addresses xenobiotic accumulation alongside lymphatic flow.	Per label direction
Lymph-Tone III (Energetix)	For patients with symptoms present for more than 3 years, heavy metal burden confirmed on Total Tox Burden, or evidence of deep degeneration. Addresses the deepest layer of lymphatic congestion. Never used simultaneously with Lymph-Tone II.	Per label direction
Methylated B12, liposomal or sublingual	Methylation support for Phase 2 liver detoxification. The liver's capacity to conjugate and eliminate debris depends on methylation capacity. Sublingual delivery bypasses the gut for patients with impaired absorption.	Per label, twice daily
ReHydration (Energetix)	Cellular hydration and electrolyte balance. Dehydration at the intracellular level is among the most consistent drainage bottlenecks in chronic illness patients.	Per label direction

All of these are available through my Fullscript dispensary, so being outside North Carolina is not a barrier to obtaining them.

For acute Lyme presentations (symptoms present less than 12 weeks), Lymph-Tone I is the appropriate drainage selection. For most chronic Lyme cases, Lymph-Tone II is standard. For patients with more than three years of symptoms or confirmed heavy metal burden, Lymph-Tone III is the clinical choice. These formulas are never used simultaneously. The drainage protocol runs for a minimum of two to three weeks before any antimicrobial support is introduced. In complex cases with multiple co-infections, confirmed mold burden, and significant chronicity, three to four weeks of drainage preparation is the appropriate standard.

Critical Timing Note: Two to three weeks of drainage before antimicrobials is how I sequence this, and in my clinical experience the preparation time is what most often separates a course a patient can finish from one they abandon. One thing it is not: a reason to delay the antibiotic treatment your physician has prescribed. If the two are in tension, the prescribed treatment wins and the drainage work runs alongside it. Timing of prescription therapy is that physician's decision, and delayed treatment is itself a documented risk factor for persistent symptoms afterward.¹²

Diet and Environmental Intervention in Lyme

Diet in Lyme disease is less about symptom management than about creating the biochemical and immunological environment in which antimicrobial treatment is most effective and recovery is most efficient. Several specific dietary principles are supported by the mechanistic understanding of how Lyme affects metabolic and immune function.

Anti-Inflammatory Foundations

The Th17-dominant immune environment of chronic Lyme disease is sustained by the same dietary factors that drive systemic inflammation in other chronic conditions: seed oils high in linoleic acid, refined sugar and high-glycemic carbohydrates, processed foods containing emulsifiers, and alcohol. Eliminating these inputs reduces the inflammatory signaling that drives cytokine production and maintains the immune polarization characteristic of active Lyme. This is not a cure for Lyme. But it removes a significant dietary driver of the inflammation that worsens every Lyme symptom.

The anti-inflammatory dietary foundation for Lyme parallels the one for Hashimoto's because the immune mechanisms overlap substantially. Remove gluten and refined carbohydrates. Replace seed oils with olive oil, avocado oil, and fat from grass-fed or wild-caught sources. Cut alcohol, which is directly hepatotoxic and impairs the liver's capacity to process the microbial debris generated during treatment. Remove processed foods containing emulsifiers and preservatives that promote gut barrier disruption.

Protein and Micronutrient Adequacy

Antimicrobial treatment for Lyme creates a significantly elevated demand for the detoxification substrates that the liver and kidneys need to process microbial debris. Glutathione, the body's primary intracellular antioxidant and detoxification molecule, is synthesized from cysteine, glycine, and glutamine, all of which are derived from dietary protein. A protein-adequate diet, with emphasis on clean animal proteins providing the full amino acid spectrum, is the dietary foundation for glutathione synthesis during active Lyme treatment.

Magnesium deficiency is near-universal in Lyme patients and has direct clinical consequences. Magnesium is required for over 300 enzymatic reactions, including ATP production, DNA repair, and the enzymatic steps in Phase 2 liver detoxification. It is depleted by chronic stress, by the inflammatory burden of active infection, and by the medications commonly prescribed for Lyme. Prioritizing magnesium-rich foods, including leafy greens, pumpkin seeds, dark chocolate, avocados, and almonds, while minimizing the dietary factors that accelerate magnesium depletion (refined sugar, alcohol, excessive caffeine), supports the metabolic foundation for recovery.

Environmental Considerations

Re-exposure prevention is a component of Lyme management that is frequently overlooked once treatment begins. Patients in tick-endemic areas who continue to have regular outdoor exposure without adequate tick prevention measures can experience repeated re-infection that undermines treatment progress. This is a reason to be systematic about protective measures, not a reason to restrict outdoor activity. That means wearing permethrin-treated clothing in wooded or grassy areas, performing full-body tick checks after outdoor exposure, and showering within two hours of potential tick exposure.

Mold burden is the most important concurrent environmental intervention for Lyme patients. The relationship between mycotoxin exposure and Lyme immune suppression is bidirectional. Lyme impairs the immune response to mold toxins, and mold toxins impair the immune response against Lyme. Patients who have both conditions, which is extremely common in clinical practice, require environmental assessment for water-damaged building exposure alongside their Lyme workup. An ERMI or HERTSMI-2 dust test of the living and working environment is the most reliable confirmation of current mold exposure. Active mold toxin burden that is not identified and addressed will consistently limit Lyme treatment outcomes.

The Lyme–Autoimmune and Neurological Connection

Borrelia burgdorferi does not just cause infection. It creates the immunological conditions in which autoimmune cascades initiate, accelerate, and persist. This is why so many chronic Lyme patients arrive with a cluster of autoimmune diagnoses that preceded, coincided with, or followed their tick-borne illness. The connection is mechanistic, not coincidental.

Understanding how *Borrelia* triggers autoimmunity changes the clinical picture in an important way: the autoimmune condition a patient develops is not determined by the organism alone. Lyme creates the terrain. The patient's genetic predispositions, prior immune exposures, and tissue vulnerabilities determine which autoimmune expression emerges from that terrain.

How *Borrelia* Triggers Autoimmunity: Five Mechanisms

The first and most well-documented is molecular mimicry. When *Borrelia* antigens share structural similarity with human tissue proteins, antibodies generated against the organism cross-react with self-tissue. This is a confirmed laboratory finding for several *Borrelia* antigens and human tissue targets, not a theoretical risk.

The second mechanism is polyclonal B-cell activation. *Borrelia* drives non-specific, broad-spectrum antibody production. This is why Lyme patients frequently show low-positive ANA, rheumatoid factor, and other autoimmune markers on standard labs even when the specific autoimmune diagnosis criteria are not met. The immune system has been pushed into non-specific antibody overproduction, and some of that production inevitably targets self-tissue.

The third mechanism is bystander activation. Chronic inflammation in tissue infected or affected by *Borrelia* activates self-reactive T cells that would otherwise remain dormant. These T cells do not require direct molecular mimicry to become pathogenic. They simply require the inflammatory environment that chronic Lyme creates.

The fourth mechanism is epitope spreading. The initial immune response against *Borrelia* antigens can widen over time to include self-antigens through a process in which the immune system, already primed and activated, begins recognizing adjacent tissue proteins as targets. This is a progressive mechanism that explains why autoimmune disease in Lyme patients can emerge months to years after the initial infection.

The fifth mechanism is sustained Th17 polarization. *Borrelia* drives a Th17-dominant immune response. Th17 cells produce IL-17 and IL-22, which drive tissue inflammation and breakdown of barrier integrity in the gut and elsewhere. Sustained Th17 dominance is the shared immune architecture of Hashimoto's thyroiditis, multiple sclerosis, rheumatoid arthritis, and lupus. Lyme sustains the same polarization that drives all of these conditions.

Documented Autoimmune Associations

Lyme arthritis is the best documented autoimmune manifestation of *Borrelia* infection. Outer surface protein A (OspA) of *Borrelia burgdorferi* shares amino acid sequence homology with lymphocyte function-associated antigen 1 (LFA-1), which is expressed on synovial T cells. The proposed consequence is a cross-reactive immune attack on joint tissue producing an arthritis pattern that persists after antibiotic eradication in a subset of patients, the condition called “post-infectious autoimmune Lyme arthritis.” The mimicry itself has been demonstrated at the single-cell level in the laboratory.⁶ That is laboratory demonstration of the mechanism, which is a genuine and specific finding, and it is not the same as proof that the mechanism accounts for the clinical course in any individual case.

The connection to demyelinating disease and multiple sclerosis has a more specific mechanistic basis than is widely appreciated. The *Borrelia burgdorferi* flagellin protein, the 41 kDa antigen that produces the most commonly positive band on the standard Western blot, generates an IgM antibody response that has been specifically demonstrated in the laboratory to cross-react with myelin basic protein.⁷ This is documented molecular mimicry between a *Borrelia* antigen and central nervous system myelin. Lyme neuroborreliosis produces white matter lesions on MRI and oligoclonal bands in cerebrospinal fluid that can be difficult to distinguish from multiple sclerosis radiographically and clinically. What I would do with that, stated as a question to raise rather than an instruction to follow: in a tick-endemic region, with a new MS diagnosis and an exposure history, tick-borne testing is worth asking your neurologist about. That is their call and their timing, not mine and not a book's. Nothing here is a reason to delay, decline, or stop a disease-modifying therapy your neurologist has recommended, and treatment decisions in MS belong entirely with the physician managing them.

The connection to Hashimoto's thyroiditis and other autoimmune thyroid conditions is proposed primarily through two pathways: the general Th17 polarization that drives thyroid lymphocytic infiltration, and a possible cross-reactivity between *Borrelia* outer surface proteins and thyroid tissue antigens. This connection is less mechanistically confirmed than Lyme arthritis or the MS-mimicry finding, but the clinical co-occurrence is real. Patients presenting with new or refractory Hashimoto's in tick-endemic areas, or whose antibodies do not normalize despite comprehensive gut repair and gluten elimination, warrant Tickborne 2.0 testing.

Lupus-like and rheumatoid arthritis-like presentations arise primarily from the polyclonal B-cell activation and Th17 polarization mechanisms. Patients with Lyme frequently show low-positive ANA and dsDNA antibodies, morning stiffness, symmetric joint involvement, and fatigue patterns that satisfy rheumatological diagnostic criteria. The distinction from primary autoimmune disease matters because the treatments diverge, and the concern raised in the literature is that immunosuppression in a patient with undiagnosed active infection may blunt the immune response containing it. Methotrexate and biologics are prescription therapies I do not prescribe, and they are mentioned here so you know the question exists and can raise it with your rheumatologist. If you are on one of them, keep taking it exactly as prescribed. The useful move is to make sure the exposure history and the tick-borne question have actually been asked, not to second-guess a prescription from a book.

The Neurodegeneration Connection: Lyme and Alzheimer's Disease

The connection between tick-borne illness and neurodegenerative disease is among the most clinically significant and underrecognized areas in this field. Dr. Dale Bredesen at the Buck Institute for Research on Aging developed the ReCODE (Reversal of Cognitive Decline) protocol, which identified multiple distinct subtypes of Alzheimer's disease. Subtype 3, which he terms "toxic" Alzheimer's, is characterized by chronic infectious burden, mycotoxin exposure, and heavy metal accumulation, a profile that overlaps precisely with the chronic Lyme and mold patient population.

The mechanistic basis for this connection is stronger than clinical observation alone. Dr. Judith Miklosy at the University of Lausanne published findings of *Borrelia burgdorferi* spirochetes and DNA detected in postmortem brain tissue of Alzheimer's disease patients, published in the *Journal of Neuroinflammation*.⁸ This is a direct detection finding, not a serology-based association.

At Harvard Medical School, researchers Robert Moir and Rudolph Tanzi published evidence in *Science Translational Medicine* that amyloid beta, long considered the pathological hallmark of Alzheimer's disease, has direct antimicrobial activity including activity against *Borrelia burgdorferi*.⁹ Under this antimicrobial protection hypothesis, amyloid deposition represents the brain's innate immune defense response to chronic microbial presence rather than a primary pathological process. If this hypothesis is correct, treating the chronic infection may be as important as addressing the amyloid itself in a subset of Alzheimer's patients.

The clinical implication is specific: patients presenting with early cognitive decline, word retrieval difficulty, brain fog with a chronic illness pattern, and a history of tick exposure or living in tick-endemic environments warrant comprehensive infectious and toxic burden assessment before a neurodegenerative diagnosis is accepted as idiopathic. Tickborne 2.0 PCR, Vibrant Mycotoxin Profile, and heavy metal testing through Total Tox Burden represent the minimum workup for any patient presenting with the Bredesen Type 3 cognitive pattern.

The evidence framing is important: Lyme-associated demyelination and the Miklosy/Bredesen neurodegenerative connections are supported by specific mechanistic findings and postmortem detection data. They are not RCT-proven causal relationships. The clinical standard for investigation is mechanistic plausibility in the setting of a treatable condition, not proof of causation. The cost of missing active Lyme or mycotoxin burden in a patient on the trajectory toward dementia is permanent. The cost of testing is low.

Your Next Steps with Root Health

Chronic tick-borne illness is one of the most complex and mismanaged conditions in conventional medicine, and the reason is missing clinical infrastructure rather than unclear science. Comprehensive testing and individualized treatment sequencing are largely absent from standard practice. The patients who recover are overwhelmingly those who find clinicians willing to test comprehensively, treat based on what testing reveals, support drainage and gut function as foundational elements of the protocol, and manage the co-infection picture rather than just the *Borrelia*.

This guide gives you the framework to understand your own clinical picture and to have more productive conversations with providers. What it cannot do is replace individualized clinical assessment and management. The specific organisms present in your case, their burden levels, the co-infection combination, your gut status, your drainage capacity, your mold burden, and whether thyroid autoimmunity is a concurrent issue all determine how this framework applies to you specifically.

Start with Comprehensive Testing

Two things come before any of it. If you were bitten in the last few days, that is a same-week conversation with a clinician about prophylaxis, not a testing project, because the intervention with the best evidence is time-limited. And if you have a rash that looks like erythema migrans, that is a clinical diagnosis in an endemic area and treatment should not wait on a blood test at all. The lab guide in this kit covers both in full.

Beyond that, the minimum comprehensive tick-borne workup is the Vibrant Tickborne 2.0 PCR, available through rootcauseunlocked.com/order-labs. For patients with fatigue that is disproportionate to their overall illness burden, add the Total Tox Burden panel to assess the mycotoxin and heavy metal co-burden. For patients with neurological or psychiatric symptoms, add the Organic Acids Test. For patients with thyroid symptoms or a history of thyroid disease, add the Hormone Zoomer. Ordering in a single draw is faster and cheaper than sequential ordering. What none of it does is confirm or exclude an infection on its own, and before you order any panel it is worth being able to answer one question about it: if this comes back positive, what specifically changes about my care.

If you want the working half of this program

This book is the reasoning. The parts you actually work from are separate documents, and they exist: **The Lyme Root Cause Kit** collects the lab interpretation guide, the diet guide and the supplement guide in one place. It is the difference between understanding why a panel matters and having the page that tells you how to read yours when it comes back.

If you would rather have everything together, the Ultimate edition bundles this book with all three. Both are at rootcauseunlocked.com/shop.

Neither is required. Everything in this book stands on its own, and the free conversation below is open to you whether you buy anything or not.

Book a Root Discovery Session

The Root Discovery Session is a free clinical conversation with me. It is a focused clinical intake, not a sales call, where your symptom history, tick exposure history, prior testing, and current clinical picture are reviewed to identify the most targeted path forward for your specific case. If you have an acute or suspected active infection, that belongs with a physician promptly, and I will tell you so rather than take the appointment. Book at rootcauseunlocked.com/discovery, or call the Garner, NC office at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Starting on your own

If you would rather start on your own, the highest-value work in this book costs nothing to begin. Protect a fixed sleep window. Pace by what you can repeat rather than by the most you can manage on a good day, especially if activity reliably produces a delayed crash. Get the deficiencies that imitate this illness ruled out, iron, B12, vitamin D, and thyroid among them. Keep a written record over months, because where testing cannot settle the question, a symptom log is genuinely one of the more useful things you can bring to an appointment. Track it at rootcauseunlocked.com/tracker.

A Personal Note

If you have been told your Lyme test is negative but your symptom history is consistent with tick-borne illness, a negative test early in the illness does not exclude the diagnosis, and that is a reason to keep the investigation open rather than to conclude anything.

If you were treated for Lyme and have not fully recovered, two things are true at once. In my clinical experience, incomplete pathogen identification and unaddressed co-infections are worth ruling out, and sometimes the answer really is in data nobody has gathered yet. And sometimes it is not. Post-treatment Lyme disease syndrome is a recognized condition with no proven treatment,¹² and anyone who tells you with confidence that one more panel or one more protocol will resolve it is selling you a certainty that does not exist. If you have been dismissed, told your labs are fine, or handed a protocol with no defined endpoint, bring all of it to the session. I will tell you honestly what I think is going on and what I would do about it, in what order, including when the answer is that this belongs with a physician rather than with me.

Dr. Adam Seay, DC · Root Health, Garner NC · rootcauseunlocked.com

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